Volume I · Chapter 3 — Healthcare Interoperability (HL7 v2, CDA/C-CDA, FHIR, SMART-on-FHIR) · Theory

> Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-I-Chapter-3-context.md` for the AI interaction log.
>
> Migrated from plan §2.6 (the template worked example).

Prerequisites: Ch 1 (Python/Pandas), Ch 2 (clinical data & code sets).

Learning outcomes — the student can:

- Explain how clinical data is **exchanged** (not just stored) and why it is the backbone of medical AI.
- **Read and parse** an HL7 v2 message and a C-CDA document into structured data.
- Model clinical data as **FHIR resources** and query a FHIR server via its **REST API** from Python.
- **Bind** data to standard terminologies (ICD/SNOMED/LOINC) and describe a **SMART-on-FHIR** integration.

Topics (theory) — 12 h:

#	Topic	h
3.1	Why interoperability matters; the data backbone of medical-AI products	1
3.2	HL7 v2.x messaging: segments, trigger	2

#	Topic	h
	events (ADT/ORM/ORU), pipe-delimited format	
3.3	HL7 CDA & C-CDA: XML clinical documents, templates, sections	2
3.4	FHIR fundamentals: resources, references, bundles, data types	3
3.5	FHIR REST API: search, read, CRUD, versioning, pagination	2
3.6	Terminology binding in FHIR (ICD/SNOMED/LOINC); profiles & conformance	1
3.7	SMART-on-FHIR & app integration; OAuth scopes, security basics	1

Datasets/tools: Synthea synthetic patients; HAPI FHIR (public sandbox or local); Python `hl7apy`, `fhir.resources`/`fhirclient`, `requests`, Pandas; sample C-CDA XML.

Assessment: quiz on standards (**20%**); lab rubric on 3a–3e (**50%**); a "clinical data → tidy DataFrame" deliverable from a FHIR bundle (**30%**).

Key decisions: FHIR vs. HL7 v2 for a new integration; **document (CDA) vs. resource (FHIR)** models; how much to normalize/terminology-map before feeding an AI pipeline.

References: plan §13 → *Interoperability standards*.

Hours: Theory **12** + Lab **16** = **28**.